FLEMING

Course Outline

Course Title:	Advanced Remote Sensing		
Course Number:	GEOM75	**Approval Date:**	2022/1/3
Course Hours:	60 hours	**Academic Year:**	2021
Academic School:	School of Environmental& Natural Resource Sciences		

Faculty:	Frank Kenny - Frank.Kenny@flemingcollege.ca
Program Co-ordinator or Equivalent:	Shawn Morgan - Shawn.Morgan@flemingcollege.ca
Dean (or Chair):	Lisa Kraemer - lisa.kraemer@flemingcollege.ca

Course Description

This advanced course further explores the topics introduced in the course Remote Sensing and Image Analysis. Students will learn the principles and operation of modern remote sensing platforms, including multispectral, thermal, Radar and LiDAR. Methods for processing and interpreting data are explored in lectures and practical skills developed through comprehensive labs and assignments. The latest feature classification techniques are also explored using various computational algorithms.

Prerequisites: GEOM 66 - Remote Sensing/Image Analysis

Corequisites: None.

Experiential Learning: Performance-Based Learning

Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Compare the application strengths and weaknesses of modern remote sensing technologies and instruments
2. Use industry standard remote sensing software
3. Design and apply customized image enhancement strategies
4. Transform and manipulate imagery to extract information
5. Conduct Multi-Spectral, Polarimetric SAR and LiDAR image classifications
6. Analyze imagery in an Image Object environment
7. Implement Photogrammetric Stereo Analysis

Learning Resources

Lecture Notes and Assigned Readings; recommended websites; on-line software help.

Suggested Text: Lillesand, Thomas M.; Kiefer, Ralph W.; Chipman, Jonathan W.; 2008. Remote Sensing and Image Interpretation, 7th ed. New York: Wiley & Sons Inc.

Stereo, Chromostereo and Anaglyph 3-D glasses - availalbe through the Fleming-Frost Book Store - Check under kits for GEOM075.

Assessment Summary

Assessment Task	Percentage
Labs	70%
Quizzes	25%
Assignments	5%

Student Success: Policies and Procedures

Mutually, faculty and learners will support and adhere to college Academic Regulations, and Student Rights and Responsibilities. The following policies and guidelines have been developed to support the learning process.

Please click on the link for information about:

- Academic Integrity (2-201A) (https://department.flemingcollege.ca/hr/attachment/7750/download)
- Accessibility for Persons with Disabilities (3-341) (https://department.flemingcollege.ca/hr/attachment/5619/download)
- Grading and Academic Standing (2-201C) (https://department.flemingcollege.ca/hr/attachment/7752/download)
- Guidelines for Professional Practice: Students and Faculty (https://flemingcollege.ca/PDF/guidelines-for-professional-practice-students-faculty.pdf)
- Student Rights and Responsibilities (5-506) (https://department.flemingcollege.ca/hr/attachment/269/download)

If you will need academic accommodations (for example if you have a learning disability, mental health condition such as anxiety or depression or if you had an IEP in high school), please contact the Accessible

Education Services (AES) department (https://department.flemingcollege.ca/aes/) to meet with a counsellor.

Alternate accessible formats of learning resources and materials will be provided, on request.

Program Standards

The **Ministry of Colleges and Universities** oversees the development and the review of standards for programs of instruction. The **Ministry of Labour Training and Skills Development** oversees the development and the review of standards for programs of instruction for Apprenticeship training in the province of Ontario. Each college is required to ensure that its programs and program delivery are consistent with these standards, and must assist students to achieve these essential outcomes.

This course contributes to Program Standards as defined by the Ministry of Colleges and Universities (MCU). Program standards apply to all similar programs of instruction offered by colleges across the province. Each program standard for a postsecondary program includes the following elements:

- **Vocational standards** (the vocationally specific learning outcomes which apply to the program of instruction in question);
- **Essential employability skills** (the essential employability skills learning outcomes which apply to all programs of instruction); and
- **General education requirement** (the requirement for general education in postsecondary programs of instruction that contribute to the development of citizens who are conscious of the diversity, complexity and richness of the human experience; and, the society in which they live and work).

Collectively, these elements outline the essential skills and knowledge that a student must reliably demonstrate in order to graduate from the program. For further information on the standards for your program, follow the MCU link (www.tcu.gov.on.ca/pepg/audiences/colleges/progstan/).

Detail Plan

Term:	2022 Winter
Faculty:	Frank Kenny - Frank.Kenny@flemingcollege.ca
Program Co-ordinator or Equivalent:	Shawn Morgan - Shawn.Morgan@flemingcollege.ca
Dean (or Chair):	Lisa Kraemer - lisa.kraemer@flemingcollege.ca

Learning Plan

Wks/Hrs Units	Topics, Resources, Learning, Activities	Learning Outcomes	Assessment
Week 1	Lecture: Course Introduction. Lab: Project 1 - Working with Image Histograms and LUTs. Lab: Assignment - Culminating Remote Sensing Technical Cartography Task	2,3,4	
Week 2	Lecture: Image Object Processing. Lab: Project 2 - Geospatial Modelling Using eCognition Image-Object Based Software	1,2,4,6	
Week 3	Lecture: Image Object Case Studies. Lab: Project 2 (continued)	1,2,4,6	
Week 4	Lecture: RADAR Remote Sensing. Lab: Project 3 - Object Processing with Machine Learning using PCI's Object Analyst	1,2,5,6	Project 2 Due (15%)
Week 5	Lecture: Image Fusion Techniques . Lab: Project 3 - (continued)	1,2,3,4,5,6	
Week 6	Lecture: Advanced Spectral Indices. Lab: Project 4 - Orthorectification and DSM (Digital Surface Model) Extraction from Legacy Aerial Photography.	2,4,7	Project 3 Due (15%)
Week 7	Lecture: LiDAR Remote Sensing. Lab: Project 4 - (continued)	2,5,7	Midterm Quiz (10%) - Tentative in Lab Period
Week 8	Independent Learning Week		
Week 9	Lecture: RADAR Interferometry. Lab: Project 4 - (continued)	2,5,7	
Week 10	Lecture: Altimeters and Oceans/Water Monitoring. Lab: Project 5 - Working with Image Parallax and Digital Elevation Models (DEM)	1,2,3,7	Project 4 Due (15%)
Week 11	Lecture: Principal Component Analysis. Lab: Project 5 - (Continued)	2,3,4,7	
Week 12	Lecture: Thermal Remote Sensing. Lab: Project 5 - (Continued)	1,2,3,7	Project 5 Due (15%)
Week 13	Lecture: Remote Sensing in Emergency Management. Lab: Project 6 - Modelling with LiDAR and Photogrammetric derived Point Clouds	1,2,4,5	Assignment #1 Due (5%)
Week 14	Lecture: Hand Held Remote Sensing Devices and Google Earth Engine. Lab: Project 6 - (Continued)	1,2,4,5	Project 6 Due (10%)
Week 15	Lab: Final Exam (D2L - Respondus Monitor).	1-7	Final Exam (15%) - Tentative in Lab Period

Assessment Requirements

Assessment Task	Date/Weeks	Course Learning Outcome	Percentage

Assessment Task	Date/Weeks	Course Learning Outcome	Percentage
Project 2: Geospatial Modelling Using eCognition Image-Object Based Software	Due Week 4	1,2,4,6	15%
Project 3: Object Processing with Machine Learning using PCI's Object Analyst	Due Week 6	1,2,5,6	15%
Midterm Quiz (D2L Respondus Monitor) - Tentative in Lab Period	Week 7	1,4,6,7	10%
Project 4 - Extracting and Visualization of Photogrammetric Parallax	Due Week 10	2,7	15%
Project 5: Working with Image Parallax and Digital Elevation Models (DEM)	Due Week 12	2,3,7	15%
Assignment 1. Culminating Remote Sensing Technical Cartography Task	Due Week 13	2,3	5%
Project 6: Modelling with LiDAR and Photogrammetric derived Point Clouds	Due Week 14	1,2,4,5	10%
Final Exam (D2L Respondus Monitor) - Tentative in Lab Period	Week 15	1-7	15%

Students are responsible for submitting assessed work as instructed on the due date and time indicated in writing, unless written arrangements have been made. Online submission time will determine late submissions. If the assessment is not submitted by the due date and time, a penalty of 10% of the assessment value per day will be assigned starting after the due date and time. This penalty applies for each 24-hour period, including weekends. After 120 hours (5 days) submissions will not be accepted.

An individual extension of the due date may be granted at the discretion of the course faculty if the student has submitted a written request. It is the student's responsibility to notify the course faculty of any problems as soon as possible. Considerations for extensions include prescribed accommodations, medical certificate or legal document. Late submissions will not be accepted under any circumstances after the assessment has been graded and distributed to the class.

Final grades in this course are assigned based on the level of academic achievement according to the Learning Plan and Assessment Requirements cited in this course outline. Faculty will not offer additional evaluation activities to individual students beyond those cited in this course outline.

Appeals of any term work evaluation must be initiated within a 5 business day period after receiving results. Appeals, as with any other academic inquiry or concern, should first be first directed to the instructor, and then, failing satisfactory resolution, as specified by the Academic Regulations Policy.

Respondus Monitor Statement

This course will use Respondus Monitor as a remote proctoring tool to support the administration of electronic examinations and assessments. Fleming College has an institutional license to use Respondus Monitor, a cloud-based service and software that captures video, audio, and other data during student assessment sessions for use in monitoring students. This data is accessible to, and may be used by, authorized individuals at the college to administer student assessments and manage the academic integrity of such assessments. The Student Terms of Use for Respondus Monitor must be agreed to by users (e.g. students) prior to each use of the Respondus Monitor, available at: https://web.respondus.com/tou-monitor-student/

By accessing and using Respondus Monitor, you agree to the collection, use, disclosure and retention of your personal information (including personal images) by Respondus Inc. in accordance with its Privacy Policy, available at: https://web.respondus.com/privacy-policy/

Questions regarding the collection of your personal information may be directed Julie Middleton, Director of Counselling and Accessible Education Services via email: julie.middleton@flemingcollege.ca

If students choose to opt out of using Respondus Monitor, they must inform their faculty member prior to the scheduled assessment, of their refusal and meet with the faculty to discuss their options.

Exemption Contact

Frank Kenny, email: frank.kenny@flemingcollege.ca

Prior Learning and Assessment and Recognition (PLAR)

PLAR uses tools to help learners reflect on, identify, articulate, and demonstrate past learning which has been acquired through study, work and other life experiences and which is not recognized through formal transfer of credit mechanisms. PLAR options include authentic assessment activities designed by faculty that may include challenge exams, portfolio presentations, interviews, and written assignments. Learners may also be encouraged and supported to design an individual documentation package that would meet the learning requirements of the course. Any student who wishes to have any prior learning acquired through life and work experience assessed, so as to translate it into a college credit, may initiate the process by applying through the Registrar's office. For more information please click on the following link: http://flemingcollege.ca/admissions/prior-learning-assessment-and-recognition

Course Specific Policies and Procedures

It is the responsibility of the student to retain this course outline for future reference. Course outlines may be required to support applications for advanced standing and credit transfer to other educational institutions, portfolio development, PLAR and accreditation with professional associations.

Synchronous sessions may be recorded. As a result, your image, voice, name, personal views and opinions, and course work may be collected under legal authority of section 2 of the Ontario Colleges of Applied Arts and Technology Act, 2002. This information will be used for the purpose of supporting student

learning. Any questions about this collection can be directed to the Privacy and Policy Officer at freedomofinformation@flemingcollege.ca or by mail to 599 Brealey Drive, Peterborough, ON K9J 7B1.

DEFINITIONS

Professor: The individual(s) who are employed by Fleming College and assigned to deliver and assess this course's approved learning outcomes in support of the program's vocational learning outcomes.

Student: Any person who is listed by the registrar as enrolled for this course while it is active.

Student Work: Any submitted assignment, exercise, presentation, product, report, code, map, idea, solution, database, web page, program, or model created by the student, in whole or in part, digital or physically printed, under the direction of the professor, where it fulfills one or more academic requirements of a class during the academic semester when the student is actively enrolled.

Group Work: Individual students may be required to work together with other students who are also enrolled in the same class.

Academic Integrity: As defined by the Academic Regulations Policy of Fleming College.

POLICIES AND PROCEDURES

Accommodation: Students who require accommodations under the Accessibility for Ontarians with Disabilities Act (AODA 2005) should ensure that their professor has received a copy of their accommodation plans.

Recording: Permission from the instructor must be obtained prior to making any live video or sound recordings of the class. Any approved recordings are to be used personally and cannot be shared and must be destroyed/deleted when the course ends.

Devices: All electronic devices (e.g. cell phones, laptops, etc.), except those approved by Accessible Education Services or professor, must be turned off or silenced for the duration of the class. Personal communication to outside entities is not permitted during the class including: Social Media (Facebook etc), Mobile Phones, Chat (SMS, Talk, Snapchat, Messenger, etc.) or any other communication unrelated to, or a distraction to the class.

Academic Integrity: Each student has the responsibility to support academic integrity. Students may be expected to work as individuals or in groups on assessments (as indicated on each assessment). No individual or group member should have direct input into another individual's or group's Student Work. Breaches of academic integrity (such as, but not limited to: plagiarism, fabrication, deception, cheating, bribery/paid services, sabotage, impersonation) will be addressed using the approved Academic Regulations Policy of Fleming College. Sharing of digital or physical copies of any Student Work for a marked assessment is considered a breach of academic integrity. Students must hand in original Student Work they (or their group) created from the current academic semester.

Lateness or Absence: Students who are late for class are a disruption and have a negative impact on the learning environment. For reasons relating to classroom management and/or laboratory/field safety, students arriving after class has commenced may be refused entry. Absences will be dealt with by the professor in accordance with Fleming College's Class Absence Operating Procedure.

Behaviour: Student behaviour in all learning environments (e.g. class, lab, field, on-line, placement, etc.) must abide by Fleming College's Student Rights and Responsibilities Policy.

Computing Resources and Software: At no time are students to contact vendors for assistance for provided software or services. All computing systems used in any part of the course are subject to the same terms and conditions under the Information Technology Appropriate Use Policy.

Public Sharing of Projects and Assignments: Any submitted Student Work may be made public for the promotion of the program and/or provided as positive examples to future enrolled students. If an individual does not wish to have their Student Work shared, it is required they submit a written request with explicit instructions on what is to be excluded to each instructor for each course.

Final Grades: Final grades in this course are assigned based on the level of academic achievement according to the Learning Plan and Assessment Requirements cited in this course outline. Faculty is not obligated offer additional evaluation activities to individual students beyond those cited in this course outline.

Modifications to the Learning Sequence in this outline may occur to facilitate the learning of the students. Notification of any changes will be posted on D2L Brightspace. Additional time will be required outside of class for completion of learning activities.